

Ali Bagheri

✉ a6bagher@uwaterloo.ca | 🌐 github.com/alibagher | 🔗 linkedin.com/in/alibagh/

Education

University of Guelph

2025 - now

Masters of Computer Science

- Thesis: Projector Camera Systems
- Graduate Courses: Technical Communication, Image Processing, AI applications and society, Human-Computer Interaction, Quantitative Research Methods

University of Waterloo

2016 - 2020

Bachelors of Computer Science (Honours)

- 4.3% program acceptance rate
- Term Dean's Honours List, Term Distinction
- Courses: Database Management, Distributed Systems, Computer Networks, Computer Architecture, Operating Systems, Algorithms, Software Design, User Interfaces, OOP Software Development, Combinatorics and Optimizations, Computational Audio, Data Structures

Work Experience

University of Guelph

Jan 2025 - now

Graduate Teaching Assistant in Operating Systems, Computer Networks, and System Analysis & Design

- Conducted weekly lab sessions to break down complex problems and help students with class concepts.
- Graded assignments, tests, and exams impartially and with attention to detail, providing positive but truthful feedback.
- Proctored exams to create an organized and respectful space for students, following university guidelines.

University of Guelph

May 2025 - Sep 2025

Full-Stack Developer

- Re-vamped back-end from Python's Flask to Django by designing APIs and structuring the project adhering to Django's documentation.
- Used DjangoQ to queue weekly calls to Google Trends API. Analyzed previous data, and after consulting with team, re-downloaded the data from last 5 years.
- Created Django admin file export/import to easily upload files that need to be processed by the backend.
- Architected using Ubuntu, Javascript, Python and PostgreSQL. Deployed website to AWS.

Ford Motor Company

Sep 2021 - Sep 2023

Software Engineer

- Automated re-basing chromium code-base with a structured, clean, and easy-to-read Python program using libraries such as Github, Repo, json, argparse, and OptionParser, resulting in 100% accuracy and improving efficiency by 30%.
- Researched and implemented V8 snapshotting; serialized application bundle into binary file to exponentially reduce load time (700%). Used V8's internal and external mechanisms, chromium, and JS (Webpack bundling, plugins, regex etc).
- Re-based chromium by resolving cherry-picked conflicts, fixing build/linking errors, and using GDB to resolve run-time errors (QNX).
- Engineered API (C++) to restrict application usage when the driver is distracted, following design patterns and coding practices.
- Developed advanced skills in git (GitHub): stash, branch, cherry-pick, remote, fetch, diff, log, show, commit, push, and pull.
- Resolved HTML/CSS/JS issues through Chrome DevTools with attention to detail resulting in smooth user experience.
- Integral member of the SDLC process with agile sprints, writing industry standard code on Linux for QNX.

Radifex

Jun 2020 - Aug 2021

Software Developer

- Created a complete Radiology software system from scratch leveraging Python/Rust for the backend, and C++, DCMTK, WASM, and Javascript for the front-end compiled on the Linux platform.
- Implemented a multi-threaded Python and Rust server. Researched running server-less WebAssembly with AWS lambda.
- Utilized object-oriented programming principals such as Observer pattern, MVC, polymorphism, encapsulation, etc.
- Identified areas for improvement in the transfer and usage of objects between C++ and JS compared to WebAssembly documentation, and took the initiative to resolve them through careful analysis and creative abstract thinking.
- Successfully combined two completely different models - multithreaded/sequential C++ and single-threaded/promise driven JS - to create a fast and light foundation for later UI, UX, optimizations and other modules.
- Improved image download and scroll speed by a factor of 3 and 2, respectively, by identifying bottlenecks.

Holloway NDT (CO-OP)

Apr 2021 - Jul 2021

Python Engineer

- Interpreted signal data from Excel sheets to present valuable pipe information using Python with 99% accuracy.
- Architected using Qt, pandas, and numpy libraries to create an OOP program that is now used on AWS back-end.

University of Waterloo - Audesse (CO-OP)

Jan 2020 - Apr 2020

Software Developer

- Transformed raw data from the car into readable and actionable information for consumers through data wrangling. Automated calculations to measure total time, idle time, gas usage, mileage with 98% accuracy using Python libraries such as time, csv, json, datetime. Final report was sent to AWS server.

Backpacker College (CO-OP)

Sep 2019 - Dec 2019

Software Consultant

- Developed full-stack programs using JavaScript resulting in efficient and accurate navigation.

Neurovative Technologies Inc (CO-OP)

Sep 2016 - Dec 2016

Junior Engineer

- Conducted R&D for company's software and hardware products by programming circuit boards using C and Arduino.
- Quality Assurance of both hardware and software products, leading to redesign and final delivery of devices.
- Use Oscilloscopes, soldering and other tools to analyze and fix circuit board issues.

Projects

Data Science

2024

- Extracted data from databases/websites using web scraping.
- Analyzed Osteoporosis data set from Kaggle, by Preparing the data (cleaning and organizing into DataFrame), Analyzing the data using GroupBy statements, and graphing the results. Analyzed using parts of the data (those under 40 years of age) due to some inconsistencies discovered in the initial analysis. Finally, the data was divided into Train-Test Split to use Machines learning algorithms such as kNN, and Random Forest to conduct further analysis.
- Analyzed Iris Flower database to distinguish the species Iris setosa, Iris virginica and Iris versicolor from each other. First, transformed the data into a Pandas DataFrame then trained kNN (k-Nearest Neighbors) algorithm to distinguish the species with accuracy score of 96%.

Email Templates

2024

- Created an end-to end system that stores email templates. Backend endpoints written in Rust and PostgreSQL - created on Docker

Hero Typer

April 2021

- A user-friendly game in Kotlin using architecture principles and design patterns to create extensible program with low coupling/high cohesion.
- Used abstraction to implement the layered, MVP and event-based architecture with OOP design patterns (factory, singleton, etc).

Distributed File System

Nov 2019

- Implemented a distributed file system using FUSE, allowing multiple clients to access and edit shared files locally.
- Designed with synchronous read and asynchronous write amongst clients using multi-thread and locks.

Go-Back-N Network Protocol

September 2019

- Used unreliable data transfer protocol (UDP) to create 100% reliable data transfer using the Go-Back-N protocol.

Operating System

Sept 2018 - Dec 2018

- Created Kernel using system calls, processes, scheduling, threads, file systems, synchronization, semaphores, and memory management to complete an Operating System.

Compiler

Spring 2018

- Created a compiler for a complete language using C++ using architectures such as assemblers, linkers etc.

More projects available upon request.

Skills

- **Systems:** C, C++, Python, WebAssembly, bash
- **Full-stack:** Python (Django, Flask, NumPy, Pandas, Matplotlib, Scikit-learn), Rust, JavaScript, Qt, WebAssembly, $\text{\LaTeX}$, CSS/HTML,
- **Tools:** Git, AWS, Jira, Visual Studio/Code, MS Office, Mercurial
- **Operating Systems:** Linux/Ubuntu, Windows, MacOS, QNX
- **Interpersonal:** Persistence, Time Management, Teamwork, Problem-Solving, Presentation, Communication, Attention to Detail, Critical Thinking, Active Listening, Patience, Strategic Planning, Professional, Leadership. Responsible

References available upon request.